

# Contents

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## Schedule

**Technologies for Artificial Intelligence** — Università degli Studi dell'Aquila First-year MSc in Computer Science, autumn 2026.

Ten lessons of three hours, every Friday from 25 September to 27 November 2026. Total: **30 hours**.

| #  | Date            | Lesson                                         | Materials                                           |
|----|-----------------|------------------------------------------------|-----------------------------------------------------|
| 1  | Fri 25 Sep 2026 | Introduction and the machine learning workflow | <a href="#">01_introduction_and_workflow</a>        |
| 2  | Fri 2 Oct 2026  | Data: exploration and preparation              | <a href="#">02_data_exploration_and_preparation</a> |
| 3  | Fri 9 Oct 2026  | Regression                                     | <a href="#">03_regression</a>                       |
| 4  | Fri 16 Oct 2026 | Classification and evaluation metrics          | <a href="#">04_classification_and_metrics</a>       |
| 5  | Fri 23 Oct 2026 | Experimental methodology and model selection   | <a href="#">05_experimental_methodology</a>         |
| 6  | Fri 30 Oct 2026 | k-NN, Naive Bayes and support vector machines  | <a href="#">06_knn_naive_bayes_svm</a>              |
| 7  | Fri 6 Nov 2026  | Trees and ensembles                            | <a href="#">07_trees_and_ensembles</a>              |
| 8  | Fri 13 Nov 2026 | Unsupervised learning                          | <a href="#">08_unsupervised_learning</a>            |
| 9  | Fri 20 Nov 2026 | Neural networks                                | <a href="#">09_neural_networks</a>                  |
| 10 | Fri 27 Nov 2026 | Convolutional networks and course synthesis    | <a href="#">10_convolutional_networks</a>           |

## Assessment dates

| Milestone               | Date                                                                                                                                                                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project brief presented | Lesson 1 — 25 September                                                                                                                                                                                                                      |
| Project topic confirmed | Lesson 4 — 16 October                                                                                                                                                                                                                        |
| Weekly exercises        | Set at the end of each lesson, discussed at the start of the next. Nothing is collected: one of the ten is drawn at the exam and you talk through your own notebook. Exercise 10 has no lesson after it, so it is worked for the exam alone. |

| Milestone                                                              | Date                                      |
|------------------------------------------------------------------------|-------------------------------------------|
| Peer review of project drafts                                          | Week of 30 November                       |
| Project submission                                                     | To be confirmed with the degree programme |
| Exam: written paper, then one of the ten exercises drawn and discussed | To be confirmed with the degree programme |

## Material release policy

Slides and notebooks are published **on the day of the lesson**. The handout, which carries the full mathematical treatment, follows **by the Monday after**, since it is study material rather than lecture support.

Students receive new material with `git pull`. The Docker image is only republished when the environment's dependencies change — see [Setup/Docker\\_Quickstart.md](#).